DSCI 554 Lab 2

Statistical Questions, Experimental Terminology, A/B Testing Communication, and Three-Way ANOVA with Blocking

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Lab Overview

This lab focuses on conceptual terminology in the design of experiments along with types of statistical questions. Moreover, it explores how we can communicate our A/B testing findings. Finally, we will work with a three-way ANOVA with blocking along with multiple comparisons.

Lab Mechanics

rubric={mechanics:5}

- Paste the URL to your GitHub repo here: **INSERT YOUR GITHUB REPO URL HERE**
- Once you finish the assignment, you must **knit** this `R` markdown to create a `.pdf` file and push everything to your GitHub repo using `git push`. You are responsible for ensuring all the figures, texts, and equations in the `.pdf` file are appropriately rendered.
- **You must submit the rendered `.pdf` file to Gradescope.**

  Heads-up: You need to have a minimum of 3 commits.

Code Quality

rubric={quality:3}

The code that you write for this assignment will be given one overall grade for code quality. Check our **code quality rubric** as a guide to what we are looking for. Also, for this course (and other MDS courses that use `R`), we are trying to follow the `tidyverse` code style. There is a guide you can refer too: http://style.tidyverse.org/

Each code question will also be assessed for code accuracy (i.e., does it do what it is supposed to do?).

Writing

rubric={writing:3}

To get the marks for this writing component, you should:

- Use proper English, spelling, and grammar throughout your submission (the non-coding parts).
- Be succinct. **This means being specific about what you want to communicate, without being superfluous.**

Check our **writing rubric** as a guide to what we are looking for.

A Note on Challenging Questions

Each lab will have a few challenging questions. These are usually low-risk questions and will contribute to maximum 5% of the lab grade. The main purpose here is to challenge yourself or dig deeper in a particular area. When you start working on labs, attempt all other questions before moving to these questions. If you are running out of time, please skip these questions.

We will be more strict with the marking of these questions. If you want to get full points in these questions, your answers need to

- be thorough, thoughtful, and well-written,
- provide convincing justification and appropriate evidence for the claims you make, and
- impress the reader of your lab with your understanding of the material, your analytical and critical reasoning skills, and your ability to think on your own.

Setup

If you fail to load any packages, you can install them and try loading the library again.

```
library(kableExtra)
library(tidyverse)
library(binom)
library(broom)
library(DescTools)
```

Exercise 1: Name that Statistical Question!

rubric={reasoning:10}

In **DSCI 552**, you were introduced to different types of statistical questions. Let us refresh our knowledge of these here and play "*name that statistical question*". For each question below, assign the answer to one of the following types of statistical question being asked:

- **Descriptive.**
- **Exploratory.**
- **Inferential.**
- **Predictive.**
- **Causal.**
- **Mechanistic.**

Question 1.1: Is wearing sunscreen associated with a decreased probability of developing skin cancer in Canada?

ANSWER:

Inferential.

Question 1.2: Is there a relationship between alcohol consumption and socioeconomic status in a given non-representative 2018 City of Vancouver dataset?

ANSWER:

Exploratory.

Question 1.3: Does a more concise Google ad lead to an increased number of visits to the advertised company's website?

ANSWER:

Causal.

Question 1.4: How do societal changes in general in human behaviour lead to a reduction in the number of COVID-19 confirmed cases?

ANSWER:

Mechanistic.

Question 1.5: Does reduced caloric intake cause weight-loss?

ANSWER:

Causal.

Question 1.6: Do tweets with GIFs get more impressions than tweets that do not?

ANSWER:

Inferential.

Question 1.7: Does including a GIF in tweets lead to more profile visits than tweets that do not include a GIF?

ANSWER:

Causal.

Question 1.8: How many mentions will my next tweet get?

ANSWER:

Predictive.

Question 1.9: How many accounts are there on Twitter today?

ANSWER:

Descriptive.

Question 1.10: Does increasing the contrast in images lead to better visual discrimination of visually impaired image content?

ANSWER:

Causal.

Exercise 2: How Would We Analyze that Data?

Q2.1. Key Terminology

rubric={reasoning:12}

It is crucial for us to have language/terminology that we can use to discuss concepts related to experimentation and causal inference. It takes time and practice for us to commit these terms and corresponding definitions to our memory to use them fluidly in practice. Let us get some more practice by matching language/terminology with their definitions.

Read the table below and assign the correct term:

- **Confounder**.
- **Randomization**.
- **Observational study**.
- **Stratification**.
- **A/B testing**.
- **Blocking**.
- **Treatment**.
- **Factor**.
- **Experimental unit**.
- **Replicate**.
- **Balanced design**.
- **Factorial design**.

Terms	Definitions
Stratification	Process of splitting the population into homogenous subgroups and sampling observational units from the population independently in each subgroup.
Factorial Design	Technique to investigate effects of several variables in one study; experimental units are assigned to all possible combinations of factors.
Confounder	A variable that is associated with both the explanatory and response variable, which can result in either the false demonstration of an association, or the masking of an actual association between the explanatory and response variable.
Blocking	Grouping experimental units (i.e., in a sample) together based on similarity.
Factor	Explanatory variable manipulated by the experimenter.
Experimental Unit	The entity/object in the sample that is assigned to a treatment and for which information is collected.
Replicate	Repetition of an experimental treatment.
Balanced Design	Equal number of experimental units for each treatment group.
Randomization	Process of randomly assigning explanatory variable(s) of interest to experimental units (e.g., patients, mice, etc.).
Treatment	A combination of factor levels.
A/B Testing	Statistically comparing a key performance indicator (conversion rate, dwell time, etc.) between two (or more) versions of a webpage/app/add to assess which one performs better.
Observational Study	A study where the researcher does not assign the primary explanatory variables of interest.

Q2.2. An Experimental Case

rubric={reasoning:10}

As part of a more extensive study to understand how aging affects learning and memory, Timbers et al. (2013) performed an experiment by using a model organism, the nematode *Caenorhabditis elegans*, to answer the following:

Question:

Does aging lead to changes in reflexive responses to mechanical stimuli?

Methods:

C. elegans were randomly assigned to be tested for their response to mechanical stimuli at either the young adult (72 hours) or old adult (120 hours) age. Response to mechanical stimuli was quantified by the distance (in millimetres) *C. elegans* crawled backwards after application of the stimulus; this is called the reversal distance.

In this study, *C. elegans* were also randomly assigned to receive a stimulus of one of the following intensities (as measured by maximum displacement): 5.5, 19.4, 30.9, and 113.8 μm. Each age of *C. elegans* was assessed for their response to all stimuli intensities. *About* 60 *C. elegans* were tested for each **treatment** group.

Your tasks:

Considering the research question and methods described above, answer the following questions:

- Was this an observational or experimental study? Explain the reasoning behind your answer **in one or two sentences**.
- What were the treatment groups for this study? If there were any factors, state their level.

  Hint: Creating a table could be useful here.

- How would you analyze the data? **Be as specific as you can (five to ten sentences).**

  Note: LLMs (such as ChatGPT) can be a valuable tool when used responsibly. Nonetheless, to develop a robust reasoning for future statistical practice both within and outside MDS, LLMs **are not permitted** to answer this case.

ANSWER:

*This is an experimental study because the researchers manipulated the assignment of the experimental unit's (*C. elegans*) levels for the factors used in the study.*

In the study, there are eight treatment groups and two factors (age and stimulus intensity). These, and their levels, are outlined in the table below:

Treatment group	Age (hours)	Stimulus intensity (μm)
A	72	5.5
B	72	19.4
C	72	30.9
D	72	113.8
E	120	5.5
F	120	19.4
G	120	30.9
H	120	113.8

I would analyze the data by performing a two-way ANOVA where reversal distance is the response variable. Age and stimulus intensity are the factors (explanatory variables). I would also investigate whether there is an interaction between the factors age and stimulus intensity. If any of the main effects (age or stimulus intensity) are significant, then post-hoc pairwise comparisons of the

treatment groups would be made to identify which group is significantly different. These pairwise comparisons would raise a multiple testing issue. Thus I would apply a Benjamini & Hochberg (BH) False Discovery Rate (FDR) adjustment (setting FDR to 5%) to the pairwise p-values to control for multiple comparisons.

Exercise 3: Communicating Findings of A/B Testing

Q3.1. The Experimental Design

rubric={reasoning:11}

The MDS marketing team designed four Facebook ads and decided to run an A/B testing to answer the following:

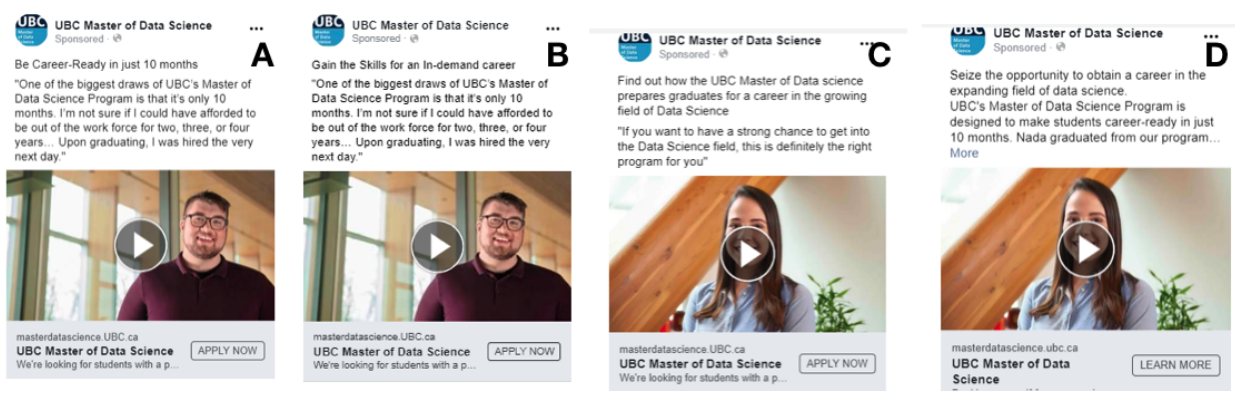
MDS Marketing Team's Questions:

1. *Which of the four ads will lead to a larger click-through rate (i.e., move the visitor from Facebook to our MDS marketing website)?*
2. *What ad features (**i.e., combinations of factor levels**) should be used in future ads?*

Methods:

The four ads are shown below (labelled `A` through `D`). These ads were randomly served to Facebook visitors for four months. The click-through rate (`CTR`) can be obtained via

$$\texttt{CTR} = \frac{\texttt{clicks}}{\texttt{impressions}}.$$



Data:

```r
# Run this code before proceeding.
MDS_ad_data <- tibble(
  ad_variant = c("A", "B", "C", "D"),
  impressions = c(11921, 6747, 2778, 29701),
  clicks = c(152, 88, 22, 450)
)
MDS_ad_data <- MDS_ad_data %>%
  mutate(CTR = round(clicks / impressions, 4))
MDS_ad_data
```

```
## # A tibble: 4 x 4
##   ad_variant impressions clicks    CTR
##   <chr>            <dbl>  <dbl>  <dbl>
## 1 A                11921    152 0.0128
## 2 B                 6747     88 0.013
## 3 C                 2778     22 0.0079
## 4 D                29701    450 0.0152
```

Your Tasks:

By looking at the images in the ads above, the two questions asked by the MDS marketing team, and the data provided; answer the following:

a. **In one or two paragraphs**, how would you statistically analyze this case (**be as specific as you can**)?
b. **In two or three paragraphs**, what would your answer be on whether your team can approach the marketing team's two questions (1) and (2) as a **full factorial** experiment?
   - Is it possible?
   - Why or why not?
   - If it is not possible, what are we only able to statistically analyze instead? Please elaborate.

   Hint: Fill out the skeleton of the table below to help you think about how this experiment can be analyzed. This table indicates the experimental factors by column indicating the treatments used by the marketing team (i.e., one row per ad used) along with the response in the column **Click-through rate (CTR)**. Also, view this **anatomy of a Facebook ad post** to help you out with the table headers).

Click-through rate (CTR)	Ad's variant label	Video	Call-to-action button (gray rectangle)	Post's text

Note: LLMs (such as ChatGPT) can be a valuable tool when used responsibly. Nonetheless, to develop a robust reasoning for future statistical practice both within and outside MDS, LLMs **are not permitted** to answer this case.

ANSWER:

I would perform a pairwise proportion test to compare the CTR for all the ad variants (or logistic regression with ad variant as the explanatory variable and a Bernoulli-type response variable). Then, I would apply a Benjamini & Hochberg (BH) False Discovery Rate (FDR) adjustment (setting FDR δ to 10%) to the p-values to control for multiple comparisons. Thus, any adjusted $p-value < 0.1$ (i.e., we want to be less strict) will be taken as a statistically significant comparison. By doing this, I could answer the marketing team's question (1) and tell them which add leads to a larger CTR.

It would be tough to answer question (2) from the marketing team. The reason for this is that many things differ between the four ads (i.e., treatments). Below, I would outline a summary of these treatments:

Click-through rate (CTR)	Ad's variant label	Video	Call-to-action button (gray rectangle)	Post's text
0.0128	A	Man	*APPLY NOW*	*Be Career-Ready in just 10-months...*
0.0130	B	Man	*APPLY NOW*	*Gain the Skills for an In-demand Career...*
0.0079	C	Woman	*APPLY NOW*	*Find out how the UBC Master of Data Science...*
0.0152	D	Woman	*LEARN MORE*	*Seize the opportunity to obtain...*

We cannot conclusively investigate the effect of the video on the ad because, although we have two treatment groups for each video, the ad text differs between all groups (and it could act as

*an unknown confounder that we cannot control for). We have a different call-to-action button (**LEARN MORE** instead of **APPLY NOW**). If all the call-to-action buttons were the same, and we had at least one additional ad variant for each video, maybe we could get some preliminary results on video. Nevertheless, these findings would have to be interpreted cautiously. Furthermore, we cannot interpret the call-to-action button's effect (even if we restrict our analysis to the woman's videos) as the text differs between the two treatment groups. Thus, either change may cause a difference in CTR between these two groups.*

The only effect we can really look at is the text's effect in the ad variants with man's videos. We can do this because we have two treatment groups where the only thing that differs is the text. Thus, we could perform a test of proportions using only these two treatment groups, with text as the explanatory variable and CTR as the response variable. Given that we would be reanalyzing the same data set that we analyzed earlier, we would also have to control for multiple comparisons using BH-FDR.

Q3.2. Exploratory Data Analysis

rubric={reasoning:4,viz:4,accuracy:2}

Now, it is time to make the statistical analysis you will present to the MDS marketing team. Create a clear and effective visualization of the data, in the form of point estimates, and include **90% confidence intervals (CIs)** (use `binom.confint()` from the package `binom` with argument `methods = "exact"`).

```
# BEGIN SOLUTION
MDS_ad_data <- mutate(MDS_ad_data, )

ci <- binom.confint(MDS_ad_data$clicks,
  MDS_ad_data$impressions,
  methods = "exact", conf.level = 0.9,
)
MDS_ad_data$lower_ci <- ci$lower
MDS_ad_data$upper_ci <- ci$upper

MDS_ad_data
```

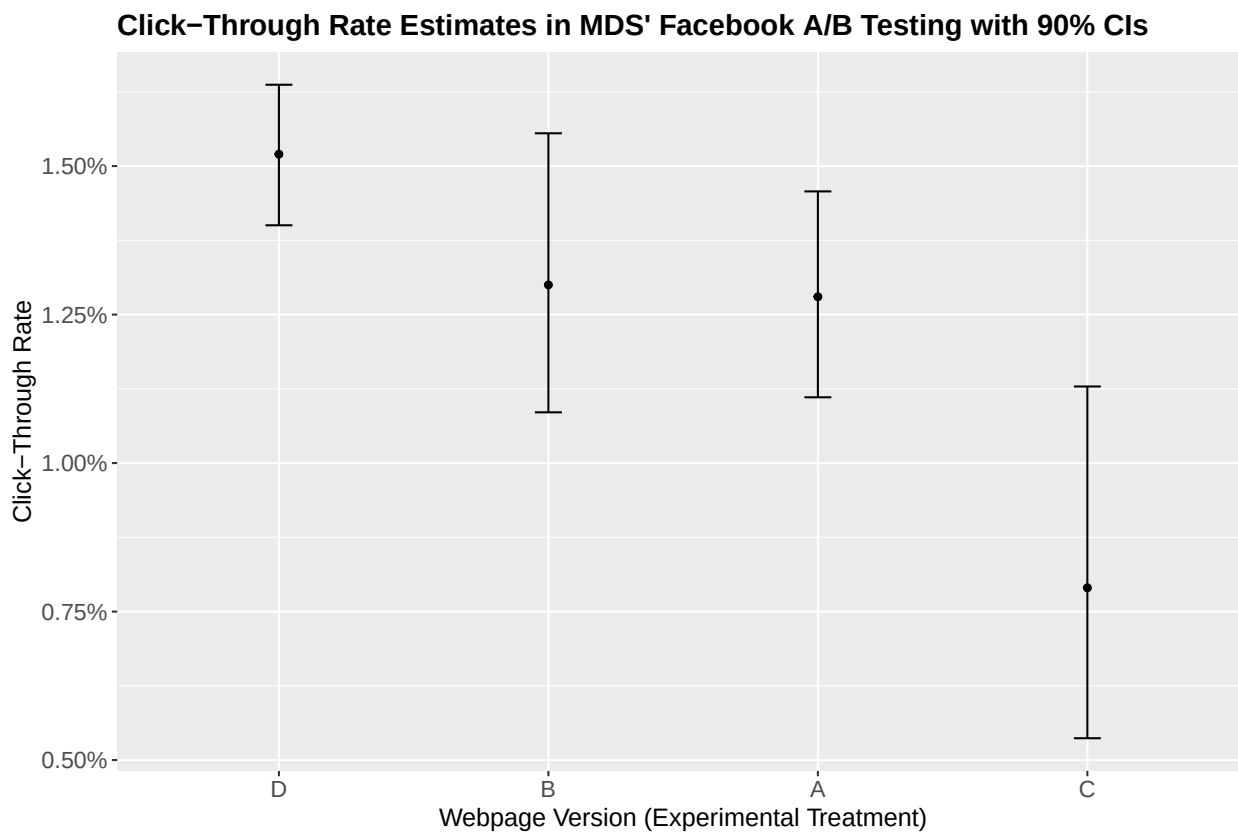
```
## # A tibble: 4 x 6
##   ad_variant impressions clicks    CTR lower_ci upper_ci
##   <chr>            <dbl>  <dbl>  <dbl>    <dbl>    <dbl>
## 1 A                11921    152 0.0128  0.0111    0.0146
## 2 B                 6747     88 0.013   0.0109    0.0156
## 3 C                 2778     22 0.0079  0.00537   0.0113
## 4 D                29701    450 0.0152  0.0140    0.0164
```

```
# Refactoring post version based on CTR in descending order.
MDS_ad_data$ad_variant <- fct_reorder(
  MDS_ad_data$ad_variant,
  desc(MDS_ad_data$CTR)
)

# Plotting as a point with 95% binomial exact confidence intervals.
CI_plot <- ggplot(MDS_ad_data, aes(ad_variant, CTR)) +
  geom_point() +
  geom_errorbar(aes(ymin = lower_ci, ymax = upper_ci), width = 0.1) +
  scale_y_continuous(labels = scales::percent) +
  xlab("Webpage Version (Experimental Treatment)") +
  ylab("Click-Through Rate") +
```

```r
  theme(
    plot.title = element_text(size = 15, face = "bold"),
    axis.text.x = element_text(size = 12, angle = 0),
    axis.text.y = element_text(size = 12, angle = 0),
    axis.title = element_text(size = 13),
    legend.text = element_text(size = 10, margin = margin(r = 0.5, unit = "cm")),
    legend.title = element_text(size = 11, face = "bold"),
    legend.position = "bottom"
  ) +
  ggtitle("Click-Through Rate Estimates in MDS' Facebook A/B Testing with 90% CIs")

CI_plot
```




```r
# END SOLUTION
```

What can we conclude from this plot? Point out any interesting finding coming from the CIs **in two or three sentences**.

ANSWER:

> *The ad's variant `D` had the largest CTR, followed by `B` and `A`. Nonetheless, we can see that the three CIs are overlapped. Simultaneously, `B` has a wider interval than `D` and `A`. It would be necessary to perform multiple pairwise comparisons to conclude which ad performs best.*

Q3.3. Statistical Analysis

rubric={reasoning:4,accuracy:3}

Statistically speaking, **in three to four sentences**, which ad performed best among all the four versions

at a significance level of 0.10?

> **Note:** The above question might sound too general given the **particular** combination of factor levels (i.e., treatments `A`, `B`, `C` and `D`) in this experiment. Hence, when answering this question, consider which ads performed better (i.e., which ones had higher CTRs) than others when making pairwise comparisons.

Conduct the corresponding pairwise tests using a tool to control for multiple comparisons **that is not too strict**. Remember to set up the argument `alternative = c("greater")` when using the corresponding pairwise comparisons function.

ANSWER:

> *We will adjust for multiple comparisons via BH-FDR. Since we already have graphical insights with CIs in **Q3.2**, we use pairwise comparisons to check what CTRs provide enough statistical evidence of being `greater` than others. Note we sort the CTRs from the smallest to the largest before conducting the hypothesis testings, so these one-sided tests make sense. Then, with $\alpha = 0.10$, for sure, we can conclude that ad `D` performed better than `C`.*
>
> *That said, even though BH-FDR indicates that `A` and `B` performed better than `C` with $\alpha = 0.10$, we might be cautious with these adjusted p-values given our previous CI results in **Q3.2** where the corresponding standalone CIs overlap. This same situation applies to `D` compared to `A`. Hence, given this context, using BH-FDR might be too lenient.*
>
> *Finally, as an additional analysis, we can compare the lenient BH-FDR results with those from more strict Bonferroni corrections (as shown below). In that regard, we can see that Bonferroni corrections indeed only indicate that ad `D` performed better than `C` with $\alpha = 0.10$. These Bonferroni conclusions align with our graphical results from **Q3.2**. Therefore, Bonferroni might be a more suitable correction method in this context.*

```r
# BEGIN SOLUTION
# Arranging CTRs from smallest to the largest
MDS_ad_data <- MDS_ad_data %>%
  arrange(CTR)

successes <- MDS_ad_data$clicks
trials <- MDS_ad_data$impressions
names(successes) <- MDS_ad_data$ad_variant
names(trials) <- MDS_ad_data$ad_variant
# END SOLUTION
```

```r
# BEGIN SOLUTION
# Using BH-FDR
multiple_comp_ad_BH <- pairwise.prop.test(successes, trials,
  alternative = c("greater"),
  p.adjust.method = "BH"
) %>%
  tidy() %>%
  unite("comparison", c(group1, group2), sep = " - ")
multiple_comp_ad_BH
```

```
## # A tibble: 6 x 2
##   comparison p.value
##   <chr>        <dbl>
## 1 A - C      0.0432
## 2 B - C      0.0432
## 3 B - A      0.459
## 4 D - C      0.00914
```

```
## 5 D - A      0.0528
## 6 D - B      0.129
```

```
# END SOLUTION
```

```
# BEGIN SOLUTION
# Using Bonferroni
multiple_comp_ad_Bonferroni <- pairwise.prop.test(successes, trials,
  alternative = c("greater"),
  p.adjust.method = "bonferroni"
) %>%
  tidy() %>%
  unite("comparison", c(group1, group2), sep = " - ")
multiple_comp_ad_Bonferroni
```

```
## # A tibble: 6 x 2
##   comparison p.value
##   <chr>        <dbl>
## 1 A - C      0.129
## 2 B - C      0.130
## 3 B - A      1
## 4 D - C      0.00914
## 5 D - A      0.211
## 6 D - B      0.645
```

```
# END SOLUTION
```

Q3.4. Experimental Conclusions

rubric={accuracy:2,reasoning:1}

Regarding the two ad's variants with the man's picture, **did the text change increase the CTR performance at a significance level of 0.10?**

ANSWER:

> *According to our results, using $\alpha = 0.10$, we do not have statistical evidence to state that both ads' CTR performance is different.*

```
# BEGIN SOLUTION
man_ads <- MDS_ad_data %>%
    filter(ad_variant == "A" | ad_variant == "B")

man_ads_results <- prop.test(man_ads$clicks, man_ads$impressions, correct = FALSE,
         alternative = c("greater")) %>%
    tidy()
man_ads_results
```

```
## # A tibble: 1 x 9
##   estimate1 estimate2 statistic p.value parameter conf.low conf.high method
##       <dbl>     <dbl>     <dbl>   <dbl>     <dbl>    <dbl>     <dbl> <chr>
## 1    0.0128    0.0130    0.0290   0.568         1 -0.00312         1 2-sample t~
## # i 1 more variable: alternative <chr>
```

```
# END SOLUTION
```

Q3.5. Communicate Your Results

rubric={reasoning:4}

In no more than 300 words, communicate the results of your **whole analysis** back to the marketing team as you would in an email. You might need to use your previous CI plot. **Recall your statistical jargon has to be adapted accordingly.**

> **Note:** LLMs (such as ChatGPT) can be a valuable tool when used responsibly. Nonetheless, to develop a robust reasoning for future statistical practice both within and outside MDS, LLMs **are not permitted** to answer this question.

ANSWER:

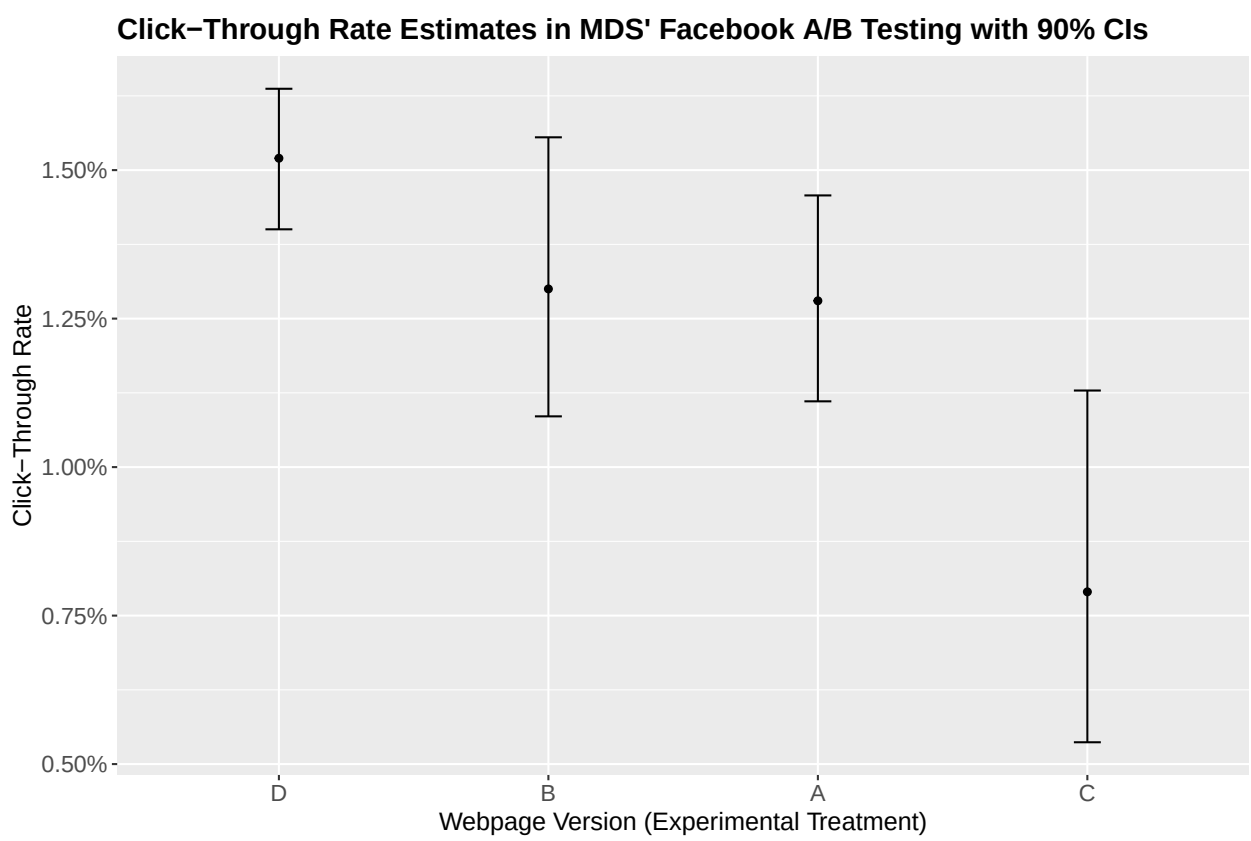
Dear MDS Marketing team,

We performed an analysis of the four ads you experimented with using the data and methodology description you provided. We found that ad `D` performed significantly better than ad `C`. The plot below indicates confidence intervals for each estimated click-through rate. Note there is overlap in the 90% confidence intervals of `D`, `A`, and `B` (unlike `D` versus `C`), which indicates that there is a chance that their population click-through rates might not be different. Therefore, we can conclude that `D` did not perform better than `A` or `B` with 90% confidence.

Given the experimental design arrangement, we cannot fully answer your second question: ***what ad features should be used in future ads?*** *We can only analyze a sub-part of this, specifically the effect of text in the ad's variants with the man's videos. When we compared the click-through rates between these groups, we found no significant difference between them. Thus, we believe that the man's videos that use the* ***APPLY NOW*** *call-to-action button and either text ("Be Career... " or "Gain the... ") could be used in the future.*

Best,

A.



(Challenging) Q3.6. Suggest a New Analysis

rubric={reasoning:4}

With no more than 300 words, write a follow-up e-mail to your Data Science team explaining how you plan to analyze the new data, which the marketing team will collect based on the improved experimental design you recommended to them.

> **Note:** LLMs (such as ChatGPT) can be a valuable tool when used responsibly. Nonetheless, to develop a robust reasoning for future statistical practice both within and outside MDS, LLMs **are not permitted** to answer this question.

ANSWER:

Dear MDS Data Science team,

I have cc'ed you all on the email to the MDS Marketing that outlines the proposed experiment I hope they will run. It will be our job to analyze the data.

Given that the response variable click-through rate is 0/1, we will use a Binomial logistic regression with ***video****,* ***call-to-action button****, and* ***post's text*** *as factors (i.e., explanatory variables).* ***Video*** *has two levels (woman and man), as does* ***call-to-action button*** *(****APPLY NOW*** *and* ***LEARN MORE****), and* ***post's text*** *has four levels (corresponding to four different texts). The Binomial logistic regression will assess if any of these factors are significant with response to ad click-through rate. We will also test whether there are any interactions between all combinations of these three factors. Significant findings from this analysis will be useful to inform us which ad features influence ad click-through rate.*

To assess whether one or more ads perform significantly better or worse than other ads, pairwise proportion tests will be run. Both the Binomial logistic regression and these pairwise proportion tests will lead us into a multiple testing problem. Thus we will use the Benjamini & Hochberg (BH) False Discovery Rate (FDR) adjustment (setting FDR to 10%).

We will also create a data visualization to communicate the point estimate (click-through rate) with 90% confidence intervals (using the binomial exact method). The ads will be ordered in the visualization via their point estimate so that the MDS Marketing team can easily see which ads performed better and which performed worse.

Best,

A.

Exercise 4: Three-Way ANOVA with Blocking

This exercise is based on data found in Lewandowski et al. (2016). It is a high-school experimental social study with three factors and a continuous response.

Questions:

- *Do pick-up lines and the subject's scent influence the receptivity for starting a relationship? What about its interaction?*
- *If so, what factor levels (or even a combination of them) increase this receptivity score?*

Methods:

This is an **unbalanced** factorial experiment with the following arrangement of three factors:

YearInSchool	Scent	PickUp
Freshman (Control)	No Spray (Control)	Direct-Direct (Control)
Freshman (Control)	No Spray (Control)	Cute-Direct
Freshman (Control)	Spray	Direct-Direct (Control)
Freshman (Control)	Spray	Cute-Direct
Sophomore	No Spray (Control)	Direct-Direct (Control)
Sophomore	No Spray (Control)	Cute-Direct
Sophomore	Spray	Direct-Direct (Control)
Sophomore	Spray	Cute-Direct
Junior	No Spray (Control)	Direct-Direct (Control)
Junior	No Spray (Control)	Cute-Direct
Junior	Spray	Direct-Direct (Control)
Junior	Spray	Cute-Direct

Our factors of interest are `Scent` and `PickUp`. Moreover, all experimental units (i.e., couples) belong to certain levels of `YearInSchool` and then randomized within each treatment of `Scent` and `PickUp`. This experiment's response is `Receptivity` to start a relationship. It is a continuous scale that goes from 1 to 5, where 1 indicates no interest in starting a social interaction and 5 indicates a strong interest to start a social interaction.

Data:

We will use the data frame `pick_up_lines_data` with the columns `Pickup`, `Scent`, `YearInSchool`, and `Receptivity`.

```r
# Run this code before proceeding.

pick_up_lines_data <- suppressMessages(read_csv("data/pick_up_lines_data.csv"))
head(pick_up_lines_data)
```

```
## # A tibble: 6 x 4
##   YearInSchool Scent    PickUp        Receptivity
##   <chr>        <chr>    <chr>               <dbl>
## 1 Sophomore    No Spray Cute-Direct          3.25
## 2 Junior       Spray    Cute-Direct          4.44
## 3 Freshman     Spray    Cute-Direct          4.44
## 4 Junior       Spray    Cute-Direct          3.92
## 5 Sophomore    No Spray Direct-Direct        3.99
## 6 Freshman     No Spray Cute-Direct          2.96
```

```r
pick_up_lines_data$YearInSchool <- as.factor(pick_up_lines_data$YearInSchool)
pick_up_lines_data$YearInSchool <- fct_relevel(
```

```r
  pick_up_lines_data$YearInSchool,
  c("Freshman", "Sophomore", "Junior")
)

pick_up_lines_data$Scent <- as.factor(pick_up_lines_data$Scent)
pick_up_lines_data$Scent <- fct_relevel(
  pick_up_lines_data$Scent,
  c("No Spray", "Spray")
)

pick_up_lines_data$PickUp <- as.factor(pick_up_lines_data$PickUp)
pick_up_lines_data$PickUp <- fct_relevel(
  pick_up_lines_data$PickUp,
  c("Direct-Direct", "Cute-Direct")
)

pick_up_lines_data %>%
  group_by(YearInSchool, Scent, PickUp) %>%
  summarize(n = n())
```

```
## `summarise()` has grouped output by 'YearInSchool', 'Scent'. You can override
## using the `.groups` argument.
```

```
## # A tibble: 12 x 4
## # Groups:   YearInSchool, Scent [6]
##    YearInSchool Scent    PickUp            n
##    <fct>        <fct>    <fct>         <int>
##  1 Freshman     No Spray Direct-Direct    10
##  2 Freshman     No Spray Cute-Direct      32
##  3 Freshman     Spray    Direct-Direct    12
##  4 Freshman     Spray    Cute-Direct      26
##  5 Sophomore    No Spray Direct-Direct    15
##  6 Sophomore    No Spray Cute-Direct      24
##  7 Sophomore    Spray    Direct-Direct    21
##  8 Sophomore    Spray    Cute-Direct      18
##  9 Junior       No Spray Direct-Direct     5
## 10 Junior       No Spray Cute-Direct       2
## 11 Junior       Spray    Direct-Direct    11
## 12 Junior       Spray    Cute-Direct      16
```

Q4.1. Explain the Experimental Design

rubric={reasoning:3}

In two or three sentences, explain what class of factor `YearInSchool` is and why.

> **Note:** When thinking about what class of factor `YearInSchool` is, do it **under an experimental framework** (NOT a basic data type such as categorical).

ANSWER:

> *`YearInSchool` is considered here as a **standalone** blocking factor. First of all, it is not a factor of primary interest in our experiment. Nonetheless, blocks allow grouping the experimental units into homogeneous groups to reduce the contribution to the experimental error of nuisance factors such as `YearInSchool`.*

Q4.2. Exploratory Data Analysis

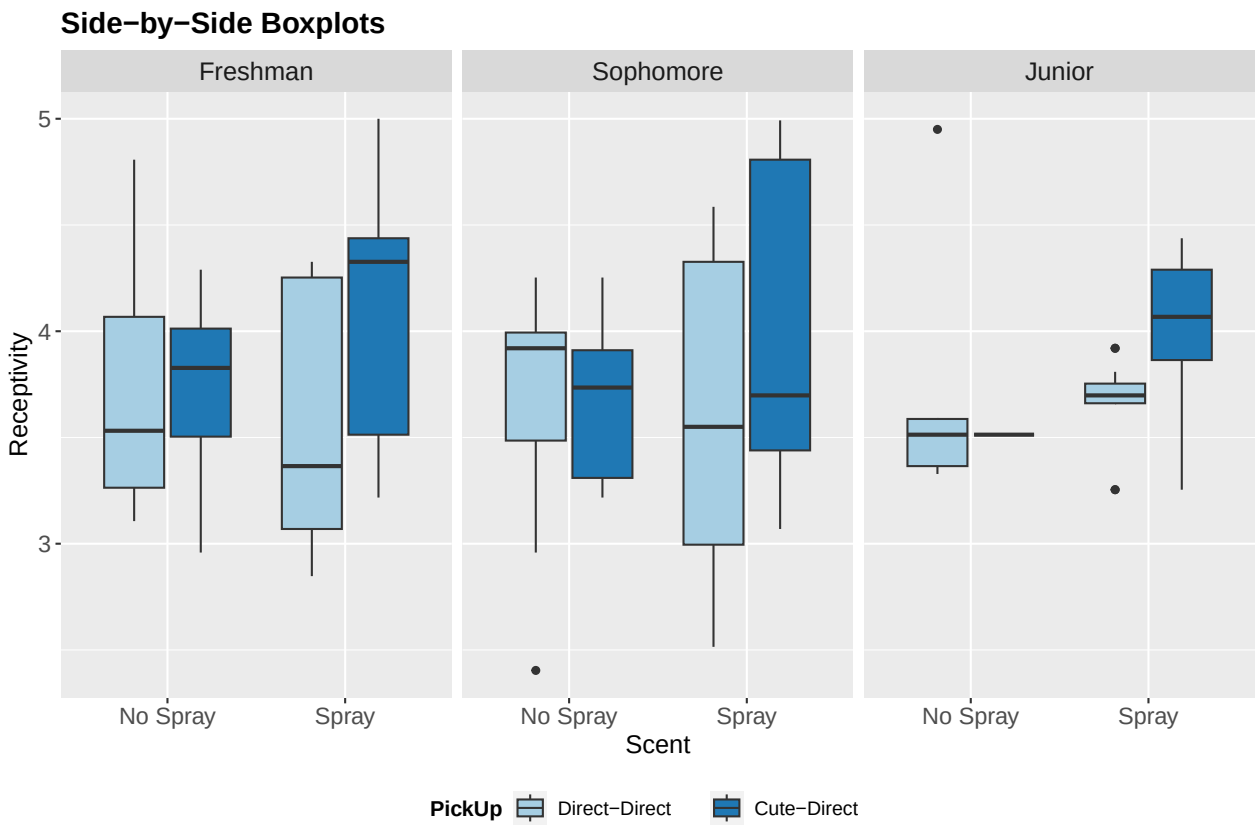
rubric={viz:4,reasoning:3}

Create a clear and effective visualization of `Receptivity` in terms of the main effects `YearInSchool`, `Scent`, and `PickUp`. Furthermore, make an interaction plot of `Scent` and `PickUp` with respect to `Receptivity`. Describe your graphical findings **in two or three sentences**.

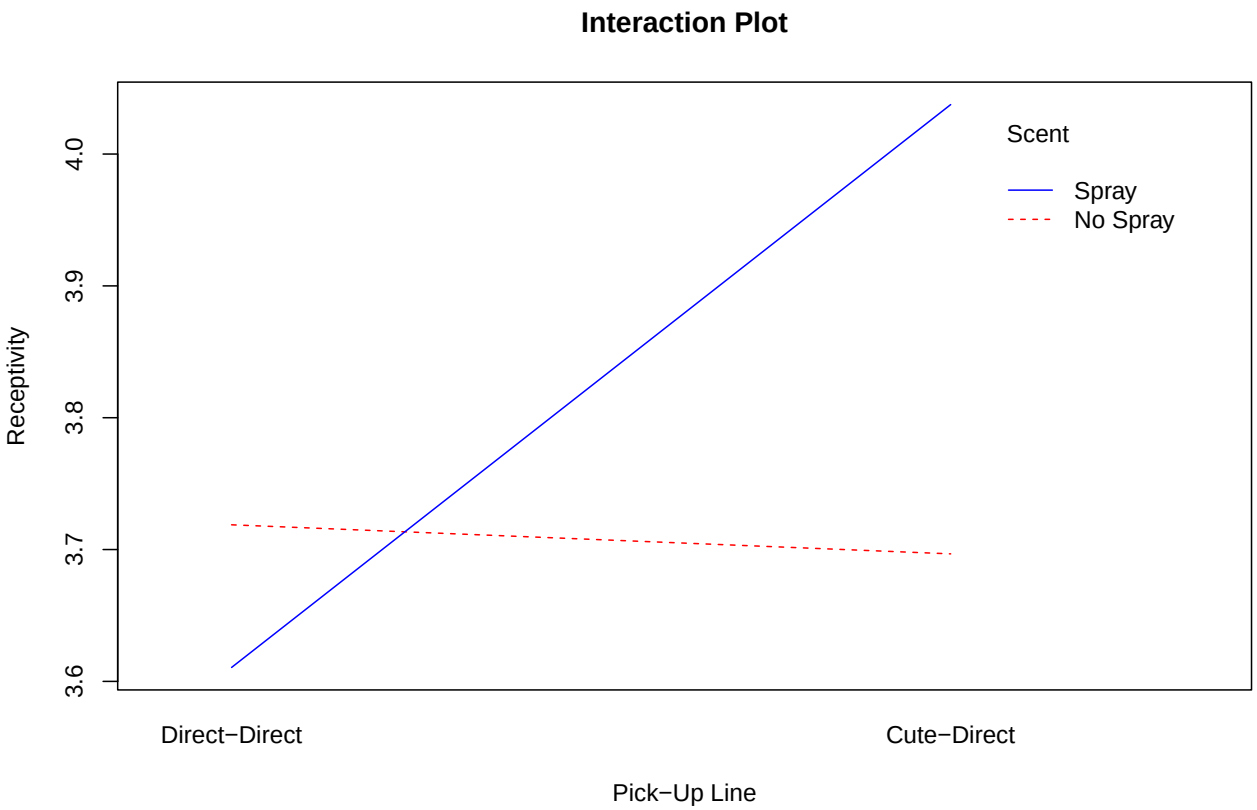
ANSWER:

> *Regarding the facets, i.e. blocking factor levels, there is no noticeable difference across them in* ***`Receptivity`****. Nonetheless, we see mixed behaviours in the levels of* ***`Scent`*** *and* ***`Pickup`****. The main effects are not providing insightful information. On the other hand, the interaction plot shows a crossed behaviour among the treatments in* ***`Scent`*** *and* ***`Pickup`****. This deserves further statistical investigation.*

```r
# BEGIN SOLUTION
pick_up_lines.boxplots <- pick_up_lines_data %>%
  ggplot(aes(x = Scent, y = Receptivity, fill = PickUp)) +
  geom_boxplot() +
  ggtitle("Side-by-Side Boxplots") +
  theme(
    plot.title = element_text(size = 15, face = "bold"),
    axis.text.x = element_text(size = 12, angle = 0),
    axis.text.y = element_text(size = 12, angle = 0),
    axis.title = element_text(size = 13),
    legend.text = element_text(size = 10, margin = margin(r = 0.5, unit = "cm")),
    legend.title = element_text(size = 11, face = "bold"),
    legend.position = "bottom",
    strip.text.x = element_text(size = 13)
  ) +
  facet_wrap(~YearInSchool) +
  scale_fill_brewer(palette = "Paired")
pick_up_lines.boxplots
```




```r
interaction.plot(
  x.factor = pick_up_lines_data$PickUp,
  trace.factor = pick_up_lines_data$Scent,
  response = pick_up_lines_data$Receptivity,
  col = c("red", "blue"),
  xlab = "Pick-Up Line", ylab = "Receptivity",
  trace.label = "Scent", main = "Interaction Plot"
)
```




```
# END SOLUTION
```

Q4.3. Data Modelling

rubric={accuracy:1}

Use `aov()` to estimate an appropriate regression model of `Scent`, `PickUp`, and `YearInSchool`:

- Use main effects and a double interaction **for your experimental factors or interest**.
- Use a standalone effect **for the blocking factor**.

Call it `pick_up_lines_anova`.

```
pick_up_lines_anova <- NULL

# BEGIN SOLUTION
pick_up_lines_anova <- aov(Receptivity ~ Scent * PickUp + YearInSchool,
  data = pick_up_lines_data
)
# END SOLUTION

pick_up_lines_anova
```

```
## Call:
##    aov(formula = Receptivity ~ Scent * PickUp + YearInSchool, data = pick_up_lines_data)
##
## Terms:
##                    Scent   PickUp YearInSchool Scent:PickUp Residuals
## Sum of Squares   1.11059  2.39552      0.18891      2.22166  48.29664
## Deg. of Freedom        1        1            2            1       186
##
```

```
## Residual standard error: 0.5095678
## Estimated effects may be unbalanced
```

Q4.4. Causal Inference

rubric={accuracy:1,reasoning:4}

Use `anova()` to conclude on the statistical significance at $\alpha = 0.05$. Moreover, **in plain words**, explain the null and alternative hypotheses for each model component (i.e., main and/or interaction effects).

ANSWER:

> *Each p-value from* ***`anova(pick_up_lines_anova)`*** *corresponds to a null hypothesis stating that all treatments* ***(associated with a given main or interaction effect)*** *are equal, versus an alternative hypothesis that at least one treatment is not equal. Using $\alpha = 0.05$, we have statistical evidence that there is a difference between both treatments, either in* ***`Scent`*** *or* ***`PickUp`****. We can also see that at least one treatment in the interaction term* ***`Scent:PickUp`*** *is statistically different.*

```r
# BEGIN SOLUTION
anova(pick_up_lines_anova) %>%
  tidy()
```

```
## # A tibble: 5 x 6
##   term            df  sumsq meansq statistic  p.value
##   <chr>        <int>  <dbl>  <dbl>     <dbl>    <dbl>
## 1 Scent            1  1.11  1.11       4.28   0.0400
## 2 PickUp           1  2.40  2.40       9.23   0.00273
## 3 YearInSchool     2  0.189 0.0945     0.364  0.696
## 4 Scent:PickUp     1  2.22  2.22       8.56   0.00387
## 5 Residuals      186 48.3   0.260     NA     NA
```

```r
# END SOLUTION
```

Q4.5. Post-hoc Tests

rubric={accuracy:1,reasoning:6}

In the design and analysis of experiments, just concluding on the previous statistical tests via `anova()` is not enough. We need to conduct **11 pairwise post-hoc tests** on the treatments by main and/or interaction effect.

The package `DescTools`, via the function `PostHocTest()`, provides the associated p-values by treatment pair. Since this is a multiple testing problem and we want to be strict, **apply a Bonferroni correction**.

These post hoc tests will allow us to fully answer the initial experimental questions on maximizing the `Receptivity` score. Make your corresponding conclusions with a significance level of 0.05. **Be as specific as possible in your reasoning.**

ANSWER:

> *First of all, it is essential to highlight that the blocking factor* ***`YearInSchool`*** *is not statistically significant. From the experiment's initial questions, we can see that we can increase the* ***`Receptivity`*** *score when using factor levels different from the control-type ones in* ***`Scent`*** *and* ***`PickUp`*** *on a main-effect framework. The estimated difference of* ***`Spray`*** *versus* ***`No Spray`*** *increases the score by 0.15, whereas* ***`Cute-Direct`*** *versus* ***`Direct-Direct`*** *increases the score by an estimated value of 0.23.*
>
> *We can see a larger score increase in interaction treatments with estimated differences larger than 0.30:* ***`Spray:Cute-Direct`*** *versus* ***`No Spray:Direct-Direct`*** *by 0.32 and* ***`Spray:Cute-Direct`***

versus `No Spray:Cute-Direct` *by 0.34. Nonetheless, the largest one is* `Spray:Cute-Direct` *versus* `Spray:Direct-Direct` *with 0.43.*

Overall, we can see that the joint use of `Spray` *with a* `Cute-Direct` *pick-up line maximizes the* `Receptivity` *score. This answers both experimental questions.*

When comparing just to baseline (`No Spray:Direct-Direct`*), only the* `Spray:Cute-Direct` *values are significant. Hence, given the results of the experiment, the treatment that maximizes the outcome of interest compared to the control treatment is the spray with cute-direct.*

```r
# BEGIN SOLUTION
PostHocTest(pick_up_lines_anova,
  method = c("bonferroni"),
  conf.level = 0.95, ordered = TRUE
)
```

```
##
##   Posthoc multiple comparisons of means : Bonferroni
##     95% family-wise confidence level
##     factor levels have been ordered
##
## $Scent
##                     diff      lwr.ci   upr.ci   pval
## Spray-No Spray 0.1526405 0.007035047 0.298246 0.0400 *
##
## $PickUp
##                                diff     lwr.ci    upr.ci   pval
## Cute-Direct-Direct-Direct 0.2286924 0.07962636 0.3777585 0.0028 **
##
## $YearInSchool
##                           diff     lwr.ci    upr.ci   pval
## Sophomore-Freshman -0.06642166 -0.2623079 0.1294646 1.0000
## Junior-Freshman    -0.01387278 -0.2658932 0.2381476 1.0000
## Junior-Sophomore    0.05254888 -0.2004333 0.3055311 1.0000
##
## $`Scent:PickUp`
##                                                    diff      lwr.ci    upr.ci
## Spray:Direct-Direct-No Spray:Direct-Direct  -0.10570067 -0.42744917 0.2160478
## No Spray:Cute-Direct-No Spray:Direct-Direct -0.01981170 -0.32541240 0.2857890
## Spray:Cute-Direct-No Spray:Direct-Direct     0.31945733  0.01559796 0.6233167
## No Spray:Cute-Direct-Spray:Direct-Direct     0.08588897 -0.18578460 0.3575625
## Spray:Cute-Direct-Spray:Direct-Direct        0.42515800  0.15544471 0.6948713
## Spray:Cute-Direct-No Spray:Cute-Direct       0.33926903  0.08903924 0.5894988
##                                                pval
## Spray:Direct-Direct-No Spray:Direct-Direct  1.00000
## No Spray:Cute-Direct-No Spray:Direct-Direct 1.00000
## Spray:Cute-Direct-No Spray:Direct-Direct    0.03353 *
## No Spray:Cute-Direct-Spray:Direct-Direct    1.00000
## Spray:Cute-Direct-Spray:Direct-Direct       0.00024 ***
## Spray:Cute-Direct-No Spray:Cute-Direct      0.00231 **
##
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# END SOLUTION
```

(Challenging) Q4.6. Types of Sum of Squares

rubric={reasoning:4}

We initially emphasized that this experiment is unbalanced. State the implications of this fact in terms of the sums of squares computation.

> **Hint:** Investigate what type of sum of squares is used in `aov()`.

ANSWER:

> *An unbalanced experiment implicates the use of different sums of squares, depending on the experimenter's objective. Traditionally, the literature indicates three types: I, II, and III. The function* `aov()` *computes the Type I, which is sequential, i.e., each factor computation is associated with a decrease in the sum of squares of the error as each factor is added to the model one at a time. Hence, depending on the computation order, these sums of squares will vary by factor, i.e.,* `aov(Receptivity ~ Scent * PickUp + YearInSchool, data = pick_up_lines_data)` *will yield different sums of squares for* `Scent` *and* `Pickup` *when compared to* `aov(Receptivity ~ PickUp * Scent + YearInSchool, data = pick_up_lines_data)`*.*

Submission

Congratulations! You are done the lab. Do not forget to:

- Knit the assignment to generate the `.pdf` file and push everything to your Github repo.
- Double check all the figures, texts, equations are rendered properly in the `.pdf` file
- **Submit the `.pdf` file to Gradescope.**

References

- Lewandowski, G., Ciarocco, N., and Strohmetz, D. (2016). *Discovering the Scientist Within: Research Methods in Psychology.* Worth Publishers.
- Timbers, T. A., Giles, A. C., Ardiel, E. L., and Kerr, R. A. (2013). Intensity discrimination deficits cause habituation changes in middle-aged caenorhabditis elegans Elsevier. *Neurobiology of Aging*, 34(2):621-631. doi:https://doi.org/10.1016/j.neurobiolaging.2012.03.016